NATIONAL CHEMISTRY CONTEST - CUBA 2020

“It always seems impossible until it’s done.”

Nelson Mandela

Name (s):______________________________ Last name:_____________________________

School:____

Municipality:______________________________ Province:______________________________

CODE:______________________________ Exam: DAY 2. 11th GRADE

Instructions

- Write all your data clearly in the indicated place (names, surnames, school, municipality, province). Do not type anything in the CODE field.
- Check that your agenda is complete. This is the agenda for the second day of the exam, 11th grade. It is made up of 2 questions (question 8, 10 items; question 9, 8 items) and a total of 4 pages.
- All necessary data, constants, conversions, and fundamental equations are included in the syllabus. You can use a scientific calculator.
- At the beginning of each question its value is indicated (out of a total of 100 points), as well as the value of each item (in marks).

Success!

Ministerio de Educación

República de Cuba

FACULTAD DE QUÍMICA

UNIVERSIDAD DE LA HABANA

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6{,}0221 \cdot 10^{23}\ mol^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases contents	$R = 8{,}3145\ J \cdot K^{-1} \cdot mol^{-1}$	The third principle of thermodynamics	$Q = \Delta E + W$
Zeroth in the Celsius scale	$273{,}15\ K$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485\ C \cdot mol^{-1}$	Nernst's equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Ionic product of water at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$	Arrhenius' equations	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p° = 100\ kPa$ $c° = 1\ mol \cdot L^{-1}$	$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$	
$1\ J = 1\ C \cdot 1\ V = 1\ kPa \cdot 1\ L = 1\ W \cdot 1\ s$		$1\ C = 1\ A \cdot 1\ s$	

PERIODIC TABLE OF THE ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 8. Nitric acid and nitrogen oxides (25 points)

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	8.10	Total	Total
8	9	2	4	3	3	2	4	6	9	50	25

Nitric acid is obtained in a large scale by the Ostwald method. In this process, ammonia is first reacted with oxygen at high temperatures and in the presence of a Pt catalyst (with small amounts of Pd or Rh) forming nitric oxide (NO). Then, nitric oxide is oxidized to nitrogen dioxide. This oxide reacts with water forming nitric acid and nitric oxide. Regenerated NO is recovered and reincorporated in the productive cycle. The process becomes more expensive because a small amount of the platinum used is transformed in PtO_2.

8.1. Write the equations corresponding to all the chemical reactions that happen in the Ostwald process.

8.2. A commercial solution of nitric acid 65% mass has a density of 1,39 $g \cdot mL^{-1}$. Calculate i) the freezing point, ii) the boiling point and iii) the vapor pressure of the solution.

8.3. The previous reaction freezes at –32 °C, boils at 121 °C and has a vapor pressure of 9,4 hPa. Why are there differences between the experimental values and the values calculated by you?

8.4. What volume of commercial solution of nitric acid is needed to prepare 1,00 L of solution with concentration 0,50 $mol \cdot L^{-}$?

Nitric oxides are generated in small amounts as byproducts of the internal combustion engines and in factories at high room temperatures. They are powerful atmospheric pollutants due to their oxidizing power and because they generate acid rain.

8.5. Calculate the pH of acid rain that contains $3,2 \cdot 10^{-5}$ $mol \cdot L^{-1}$ of nitric acid.

Nitrogen dioxide exists in equilibrium with the N_2O_4 dimer, dinitrogen tetroxide. This oxide is used as rocket fuel.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

8.6. Draw the Lewis structure of N_2O_4.

8.7. The conversion of NO_2 to N_2O_4 is favored at low temperatures. Why?

To study the balance between these oxides, 2.00 mol of NO_2 were introduced into a 5.00 L rigid container, previously evacuated. The container was cooled with ice to 0 °C.

8.8. Estimate the approximate total pressure (in kPa) inside the container at 0 °C. Assume that only N_2O_4 dimer is present under these conditions.

8.9. At what temperature is it true that Kp° = 1.00 for the chemical equilibrium analyzed?

8.10. Calculate the total pressure (in kPa) inside the container at the above temperature. If you could not calculate the value of the temperature in 8.8, use the value T = 50 °C.

$K_{cr}(H_2O) = 1,86\ °C \cdot mol^{-1} \cdot kg^{-1}$

$K_{eb}(H_2O) = 0,52\ °C \cdot mol^{-1} \cdot kg^{-1}$

$p°(H_2O) = 31,7$ hPa $R = 0,08205\ atm \cdot L \cdot K^{-1} \cdot mol^{-1}$

Substance	NO_2	N_2O_4
$\Delta H_f°$ ($kJ \cdot mol^{-1}$)	33,18	9,16
$S°$ ($J \cdot K^{-1} \cdot mol^{-1}$)	240,06	304,29

Problem 9. Apatite (25 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	Total	Total
3	5	8	5	3	4	15	7	50	25

Among the minerals known as apatites, those with the simplest composition have the formula $Ca_5(PO_4)_3X$; the X^- anion can be Cl^- (chloroapatite), F^- (fluoroapatite) or OH^- (hydroxyapatite). Apatites are the main sources of phosphate, and therefore are essential in the manufacture of mineral fertilizers and obtaining phosphorus. Hydroxyapatite is present in the bones and teeth of the human body. It constitutes about 60-70% of bone tissue, to which it gives great mechanical resistance; and contains 99% of the calcium and 80% of the total phosphorus. Hydroxyapatite is a mineral that is slightly soluble in water, so it has a very small solubility product.

$$\mathrm{Ca_5(PO_4)_3OH(s)} \rightleftharpoons \mathrm{5Ca^{2+}(ac) + 3PO_4{}^{3-}(ac) + OH^-(ac)} \qquad K_{ps} = 3{,}0 \cdot 10^{-58}$$

9.1. Hydroxyapatite is dissolved by the action of mineral acids such as hydrochloric acid. State the ionic reaction that occurs; suppose that $H_2PO_4^-$ is formed.

Hydrolysis of dental enamel hydroxyapatite causes the formation of caries. This process occurs when the bacteria present in the mouth metabolize the sugars in the remains of food and transform them into monoprotic acids such as acetic and lactic. When the pH of saliva is less than 5.5, hydroxyapatite begins to dissolve and tooth decay occurs. For this reason, it is necessary to maintain adequate oral hygiene.

9.2. Calculate the molar concentration of lactic acid required to bring the pH of saliva to 5.5.

9.3. Show that the solubility S (in $mol \cdot L^{-1}$) of hydroxyapatite in water, as a function of pH and the corresponding equilibrium constants, obeys the following expression:

$$S = \sqrt[8]{\frac{K_{ps} \cdot 10^{-pH}}{84375 \cdot \alpha_3{}^3 \cdot K_w}} \qquad \alpha_3 = \frac{[PO_4{}^{3-}]}{c(PO_4{}^{3-})_{total}} = \frac{K_{a1}K_{a2}K_{a3}}{[H^+]^3 + K_{a1}[H^+]^2 + K_{a1}K_{a2}[H^+] + K_{a1}K_{a2}K_{a3}}$$

Note: You do not have to prove the expression for α_3.

9.4. Calculate the solubility (in $mol \cdot L^{-1}$) of hydroxyapatite in water at pH = 5.5.

Many commercial mouthwashes and toothpastes add fluoride to their formulation to help prevent cavities. However, excess fluoride is harmful, so ingestion of these preparations should be avoided and use in young children should be controlled. The fluoride ion is capable of stopping tooth decay because it allows the formation of fluoroapatite, which is less soluble ($K_{ps} = 7.1 \cdot 10^{-61}$) than hydroxyapatite, according to the reaction

$$\mathrm{Ca_5(PO_4)_3OH(s) + F^-(ac)} \rightleftharpoons \mathrm{Ca_5(PO_4)_3F(s) + OH^-(ac)}$$

9.5. Calculate the equilibrium constant for the above reaction.

9.6. Calculate the concentration of fluoride ions required for fluoroapatite to form at pH 5.5. If you could not solve part 9.5, consider K = 500.

9.7. Calculate the solubility of fluoroapatite in water at pH = 5.5.

9.8. Is there solid calcium fluoride ($K_{ps} = 3.6 \cdot 10^{-11}$) in equilibrium with the fluorapatite under the above conditions? If you were unable to solve part 9.7, consider $S = 1.00 \cdot 10^{-5}$ $mol \cdot L^{-1}$.

acid	acetic	lactic	HF	H_3PO_4
pK_a	4,76	3,86	3,17	2,12; 7,21; 12,67

$pX \equiv -\log X$